

TEEL 1210 – Electrician Apprentice IIA (3 Credits)

Course Description

The Electrician Apprentice IIA course discusses single-phase and three-phase alternating current (AC) power systems, inductance, capacitance, reactance, power factor, and power correction. In this course, students will begin a more comprehensive analysis of National Electrical Code (NEC) requirements and calculations. They will explore the NEC requirements for wiring methods and installations of electrical systems as well as electrical safety in the use of energized equipment.

Course Objectives

- Show proficiency in calculating properties of an AC circuit.
- Demonstrate proper use of hand tools and electrical equipment in practice live applications
- Demonstrate proficiency in applying and calculating the sizing of Branch circuits, feeders, services, and load calculations

Course Outline

- Year IIA

Textbook & Reading Materials

NFPA 70: NEC National Electrical Code 2026 (ISBN: 9781455932207)

NEC 2026 Colored Fast Tabs (ISBN: 9781622707249)

Ugly's Electrical References, 2026 Edition (ISBN: 9781284315752)

Electrical Safety: A Practical Guide to OSHA and NFPA 70E (ISBN: 9780826935977)

Delmar's Standard Book of Electricity (ISBN: 9781337900348)

Electrical Systems Based on the 2023 NEC (ISBN: 9780826920638)

Assignments & Assessments

Orientation

Orientation Acknowledgement

Electrical Safety: OSHA and NFPA 70E CH 3

UNIT 6 Series Circuits

Reading

DEL Unit 6 Series Circuits Due 9-9

UNIT 7 Parallel Circuits

Reading

DEL Unit 7 Parallel Circuits Due 9-9

UNIT 8 Combination Circuits

Reading

DEL Unit 8 Combination Circuits Due 9-9

UNIT 14 Basic Trigonometry and Vectors

Reading

DEL Unit 14 Basic Trigonometry and Vectors Due 9-16

UNIT 15 Alternating Current

Reading

DEL Unit 15 Alternating Current Due 9-16

UNIT 16 Inductance in AC Circuits

Reading

DEL Unit 16 Inductance in AC Circuits Due 9-23

UNIT 17 Resistive-Inductive Series Circuits

Reading

DEL Unit 17 Resistive-Inductive Series Circuits Due 9-30

UNIT 18 Resistive-Inductive Parallel Circuits

Reading

DEL Unit 18 Resistive-Inductive Parallel Circuits Due 9-30

UNIT 19 Capacitors

Reading

DEL Unit 19 Capacitors 10-7

UNIT 20 Capacitance in AC Circuits

Reading

DEL Unit 20 Capacitance in AC Circuit Due 10-7

Midterm Exam

UNIT 21 Resistive-Capacitive Series Circuits

Reading

DEL Unit 21 Resistive-Capacitive Series Circuits Due 11-11

UNIT 22 Resistive-Capacitive Parallel Circuits

Reading

DEL Unit 22 Resistive-Capacitive Parallel Circuits Due 11-11

UNIT 23 Resistive-Inductive-Capacitive Series Circuits

Reading

DEL Unit 23 Resistive-Inductive-Capacitive Series Circuits 11-18

18

UNIT 24 Resistive-Inductive-Capacitive Parallel Circuits

Reading

DEL Unit 24 Resistive-Inductive-Capacitive Parallel Circuits Due 11-18

UNIT 10 Using Wire Tables and Determining Conductor Sizes

Reading

DEL Unit 10 Using Wire Tables and Determining Conductor Sizes Due 11-25

UNIT 26 Three-Phase Circuits

Reading

DEL Unit 26 Three-Phase Power Due 12-9

eTextbook: Illustrated Guide to the National Electrical Code

IGNEC Unit 8 Load Calculations Due 12-9

ES Chapter 2-Branch Circuits and Feeders Due 11-16-23

ES Chapter 3-Dwelling Load Calculations Due 12-1-22

ES Chapter 4-Services Due 12-1-22

ES Chapter 5-Conductors and Overcurrent Protection

Final Practice Exam

ELEC 002A Final Exam

Bell Quiz 1 9-22-22

Bell Quiz 2 10-6-22

Bell Quiz 3 10-13-22

Bell Quiz 4 11-10-22

Bell Quiz 5 11-17-22

End of Course Survey

1_found_base_plan.pdf

2_first_floor plan.pdf

3_south_elev wind_door _sch.pdf

4_east_elev int_elev.pdf

S_north_elev_plot_plan.pdf

6_west_elev_elec_sch.pdf

7_cross_section_a-a.pdf

8_basement_elec_plan.pdf

9_first_floor elec_plan.pdf

basement_electrical.pdf

first_floor _electrical.pdf

Subject to change. Please consult your Canvas course for the most current instructions and updates.

Classroom Hours

Start: 08-24-2026

End: 12-14-2026

Mo, Th

6:00 PM - 9:00 PM

For a full list of course hours visit: [Course Schedule](#)

Instructor Contact Information

Jeremy Leavitt — jleavitt@stech.edu

Office Hours: By appointment

Email is the preferred method of communication; you will receive a response within 24 hours during regular business hours.

Canvas Information

Canvas is where course content, grades, and communication will reside for this course.

- Canvas URL: stech.instructure.com
 - For Canvas passwords or any other computer-related technical support contact Student Services.
 - For Regular Hours and Weekdays call (435) 586 - 2899
 - For After Hours and Weekends call (435) 865 - 3929 (Leave a message if no response)
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Policies

Course Grading

You will also be required to complete the course, exams and homework, with 75% or higher accuracy for credit and DOPL licensure points.

High School Power School Grades: Quarter student grades will be determined by student progress percentage. Faculty will use the higher percentage of either 1) quarter progress, or 2) cumulative progress for the current training plan year.

Grade Scale: The following grading scale will be used to determine a letter grade.

A : 94 - 100%	A- : 90 - 93%	B+ : 87 - 89%	B : 83 - 86%
B- : 80 - 82%	C+ : 77 - 79%	C : 73 - 76%	C- : 70 - 72%
D+ : 67 - 69%	D : 63 - 66%	D- : 60 - 62%	F : 0 - 59%

Course Policies

Cell phones for many have become a distraction. When you are in class or lab we encourage you to keep your cell phones put away in a secure location. If you use ear buds we ask that you only use one so you can still hear the things going on around you. If you are using your phone for things other than school related items, instructors will ask you to put them away. Please follow the direction of your instructors. Those who have been asked to refrain from using your cell phone and fail to do so will be asked to meet with the Director of Transportation, and student services will be notified.

Additional Information

InformaCast Statement: Southwest Tech uses InformaCast to ensure the safety and well-being of our students. In times of emergency, such as weather closures and delays, this app allows us to promptly deliver notifications directly to your mobile devices. To stay informed and receive real-time updates, we encourage all students to sign up for notifications. Your safety is our priority, and staying connected ensures a swift response to any unforeseen circumstances. More information and directions for signing up are available at: <https://stech.edu/emergency-notifications/>

Internet Acceptable Use Policy: The student is expected to review and follow the Southwest Technical College Internet Safety Policy at: <https://stech.edu/students/policies/>

Student Code of Conduct Policy: The student is expected to review and follow the Southwest Technical College Student Code of Conduct Policy at: <https://stech.edu/students/policies/>

Accommodations: Students with medical, psychological, learning, or other disabilities desiring accommodations or services under ADA, must contact the Student Services Office. Student Services determines eligibility for and authorizes the provision of these accommodations and services. Students must voluntarily disclose that they have a disability, request an accommodation, and provide documentation of their disability. Students with disabilities may apply for accommodations, based on an eligible disability, through the Student Services office located at 757 W. 800 S., Cedar City, UT 84720, and by phone at (435) 586-2899. No diagnostic services are currently available through Southwest Technical College.

Safety and Building Maintenance: The College has developed and follows a variety of plans to ensure the safe and effective operation of its facilities and programs. The following plans are available online: 1) Facilities Operations and Maintenance Plan; 2) Technical Infrastructure Plan; and 3) Health and Safety Plan.

Withdrawals and Refunds: Please refer to the Southwest Technical College Refund Policy at: <https://stech.edu/students/policies/>

Any high school or adult student, who declares a technical training objective is eligible for admission at Southwest Technical College (Southwest Tech). Program-specific admissions requirements may exist and will be listed on the Southwest Tech website. A high school diploma or equivalent is not required for admission but is mandatory for students seeking Title IV Federal Financial Aid.

Non-Discriminatory Policy: Southwest Technical College affirms its commitment to promote the goals of fairness and equity in all aspects of the educational enterprise, and bases its policies on the idea of global human dignity.

Southwest Tech is committed to a policy of nondiscrimination. No otherwise qualified person may be excluded from participation in or be subjected to discrimination in any course, program or activity because of race, age, color, religion, sex, pregnancy, national origin or disability. Southwest Technical College does not discriminate on the basis of sex in the education programs or activities that it operates, as required by Title IX and 34 CFR part 106. The requirement not to discriminate in education programs or activities extends to admission and employment. Inquiries about Title IX and its regulations to STECH may be referred to the Title IX Coordinator, to the Department of Education, and/or to the Office for Civil rights.

If you believe you have experienced discrimination or harassment on our campus, please contact the Title IX Coordinator, Cory Estes: cestes@stech.edu, (435) 865-3938.

For special accommodations, please contact the ADA Coordinator, Cyndie Tracy: ctracy@stech.edu, (435) 865-3944.

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